BITS Pilani - M.Tech (AI/ML)

Machine Learning | Assignment 2

Student: Shiv Nandan Yadav | Student ID: 2026ak05694

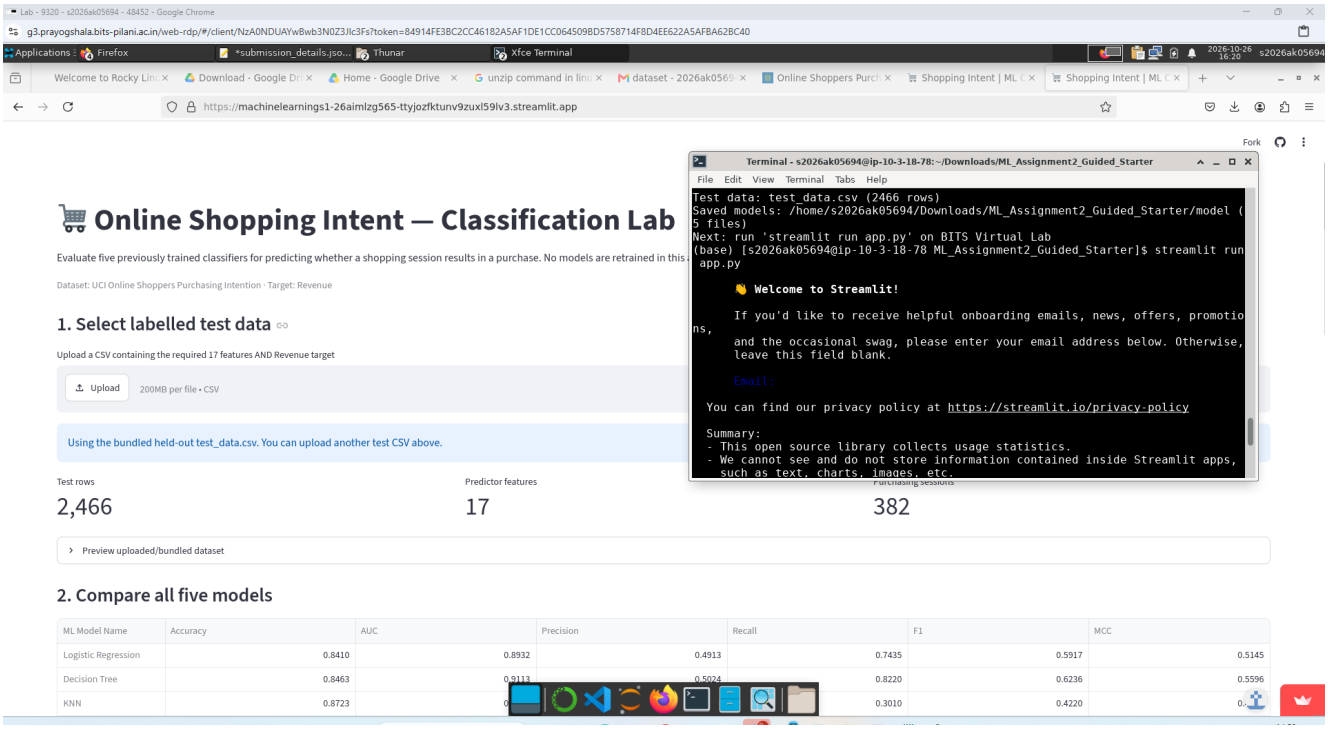
1. GitHub repository link

https://github.com/ShivBITS-Pillani/Machine_Learning_S1-26_AIMLZG565.git

2. Live Streamlit application link

https://machinelearnings1-26aimlzg565-ttyjozfktunv9zuxl59lv3.streamlit.app/

3. Evidence: BITS Virtual Lab execution screenshot



Evidence screen shot.

4. Complete GitHub README.md content

Online Shopping Purchase Intent - Machine Learning Assignment 2

This repository supports a reproducible comparison of five classifiers.

All final metrics and observations must be validated on BITS Virtual Lab.

a. Problem statement

Predict whether a customer browsing an e-commerce website will finish the session with a purchase, using 17 recorded session features (Revenue target). The five classifiers are fitted to a shared stratified training split and are compared on an untouched held-out test split.

b. Dataset description

- Source: UCI Machine Learning Repository, dataset ID 468.
- Dataset name: Online Shoppers Purchasing Intention Dataset.
- Source URL: https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset
- Instances: 12,330 shopping sessions.
- Source predictor features: 17, with a binary target named Revenue.
- No-purchase class: 10,422 sessions; purchase class: 1,908 sessions.
- Distribution is imbalanced; Accuracy alone can be misleading.
- License: CC BY 4.0; credit the dataset's original contributors.

c. GitHub Repository Link

https://github.com/ShivBITS-Pillani/Machine_Learning_S1-26_AIMLZG565.git

d. Models used - Comparison table

The following results were calculated on the same untouched labelled held-out test split.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8410	0.8932	0.4913	0.7435	0.5917	0.5145
Decision Tree	0.8463	0.9113	0.5024	0.8220	0.6236	0.5596
KNN	0.8723	0.8285	0.7055	0.3010	0.4220	0.4049
Naive Bayes	0.2887	0.7571	0.1752	0.9686	0.2967	0.1369
Support Vector Machine	0.8654	0.8979	0.5477	0.7513	0.6336	0.5639

Model performance observations

ML Model Name	Observation about model performance
Logistic Regression	Logistic Regression achieved 84.10% Accuracy, 0.8932 AUC, 0.4913 Precision, 0.7435 Recall and 0.5917 F1 Score. The model identifies approximately 74% of actual purchasers but produces some false purchase predictions, as reflected in its lower Precision. Class balancing helps prioritize the minority purchase class.
Decision Tree	Decision Tree achieved 84.63% Accuracy, the highest AUC of 0.9113 and 82.20% Recall. Its F1 Score of 0.6236 and MCC of 0.5596 indicate strong overall performance. The model detects more actual purchasers than SVM but has lower Precision. The configured depth and minimum leaf size help control model complexity, although training performance would be needed to assess overfitting.
KNN	KNN achieved the highest Accuracy of 87.23% and Precision of 0.7055. However, Recall was only 0.3010, with an F1 Score of 0.4220 and MCC of 0.4049. This indicates that KNN makes relatively accurate positive predictions but misses many actual purchasing sessions. Its high Accuracy is therefore misleading for this imbalanced classification problem.
Naive Bayes	Gaussian Naive Bayes achieved the highest Recall of 0.9686 but the lowest Accuracy of 28.87%. Its Precision of 0.1752, F1 Score of 0.2967 and MCC of 0.1369 indicate a large number of false positive predictions. The independence assumption and probability estimates may contribute to its poor classification performance on this dataset.
Support Vector Machine	SVM achieved 86.54% Accuracy, 0.8979 AUC, 0.5477 Precision and 0.7513 Recall. It obtained the highest F1 Score of 0.6336 and MCC of 0.5639 among the five models. The RBF kernel allows nonlinear decision boundaries, and the results show a comparatively strong balance between detecting actual purchasers and limiting false positive predictions.
Overall winner for your dataset?	Support Vector Machine is selected as the overall winner because it achieved the highest F1 Score (0.6336) and MCC (0.5639), along with 86.54% Accuracy and 0.8979 AUC. These metrics make it a strong balanced choice for the imbalanced online shopping dataset. Decision Tree remains a competitive alternative because it achieved the highest AUC (0.9113) and better Recall (0.8220), which may be preferable when detecting more purchasers is the primary objective.

Training and deployment methodology

- Train/test split: 80%/20%, stratified by Revenue, random_state=42.

- Preprocessing: median imputation and scaling for numeric predictors;

categorical imputation and one-hot encoding for categorical predictors.

- All preprocessing is fitted on training data within each saved Pipeline.

- Five classifiers: Logistic Regression, Decision Tree, KNN, Gaussian Naive

Bayes and RBF Support Vector Machine.

- Evaluation: Accuracy, ROC AUC, Precision, Recall, F1 and MCC.

- Live app loads the trained Pipelines without fitting on uploaded test data.

Reproduce in BITS Virtual Lab

1. Install packages: python -m pip install -r requirements.txt

2. Get UCI data: python download_dataset.py (or manual download).

3. Train and save test set: python train_models.py

4. Test app: streamlit run app.py

5. Upload test_data.csv and compare all models.

6. Review and write your own model observations.

7. Update submission_details.json and run python prepare_submission.py.

Files

- train_models.py: model fitting and saved hold-out test set.

- common.py: shared schema and evaluation metrics.

- app.py: interactive Streamlit evaluation, no retraining.

- model/*.joblib: five fitted preprocessing + model Pipelines (after training).

- test_data.csv: held-out labelled test data (after training).

- model_comparison.csv: six-metric results table (after training).

- requirements.txt: application dependencies.

- prepare_submission.py: build final README and one-PDF submission.

Attribution

Sakar, C. and Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset.

UCI Machine Learning Repository. DOI: 10.24432/C5F88Q.